# Before class: a race between two phases

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

## 1 · Read first (5 min)

### Like dissolves like, again

Polar molecules attract polar molecules; nonpolar ones mix with nonpolar ones (Chapter 1). Groups such as –OH, $–NH_2$ and –COOH make a molecule more polar; long hydrocarbon chains make it less polar. Chromatography uses exactly this idea.

### A race between two phases

In chromatography a **mobile phase** (a liquid or a gas) flows over a **stationary phase** (often a solid such as silica). A compound that sticks to the stationary phase moves slowly; one that prefers the mobile phase moves quickly. So a mixture spreads out into separate spots or peaks.

### Why it matters in pharmacy

Every batch of medicine is tested: is it the right drug, is it pure, and is the dose correct? Hospitals measure drug levels in blood. Regulators look for hidden drugs in 'herbal' products. Most of these tests use **TLC**, **HPLC** or **GC**.

## 2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

Which molecule is the most polar?

A. methanol, $CH_3OH$

B. hexane, $C_6H_{14}$

C. methane, $CH_4$

D. benzene, $C_6H_6$

Q2 Recap

Which group can form hydrogen bonds?

A. –OH

B. $–CH_3$

C. –C≡C–

D. –Cl

Q3 Recap

Above its pI, an amino acid carries a net:

A. negative charge

B. positive charge

C. zero charge

D. it depends on temperature

Q4 CH8-LO1

In chromatography, the phase that moves is called the:

A. mobile phase

B. stationary phase

C. solid phase

D. fixed phase

Q5 CH8-LO1

A compound that sticks strongly to the stationary phase travels faster.

True / False

Q6 CH8-LO2

Silica gel is:

A. polar
B. nonpolar
C. a gas
D. an ion

Q7 CH8-LO3

TLC stands for:

A. thin layer chromatography
B. total liquid chromatography
C. thermal layer chemistry
D. test for liquid content

Q8 CH8-LO4

In gas chromatography, the sample must be:

A. able to become a vapour without breaking down
B. a solid salt
C. a protein
D. coloured

Q9 CH8-LO5

Which question does a pharmacy quality-control lab NOT need chromatography for?

A. the price of the tablet

B. the identity of the drug

C. the amount of drug

D. the impurities present

Q10 CH8-LO1

Have you seen ink or food colouring spread out on wet paper or tissue? Describe what happened and suggest why the colours separated.

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PHD125 Pharmaceutical Organic Chemistry · Application of Chromatographic Methods in Pharmaceutical Analysis

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## Answer key

### 2 · Readiness check

Q1 Recap

Which molecule is the most polar?

**A. methanol, $CH_3OH$**

B. hexane, $C_6H_{14}$

C. methane, $CH_4$

D. benzene, $C_6H_6$

The –OH group makes it polar and able to hydrogen bond.

Q2 Recap

Which group can form hydrogen bonds?

**A. –OH**

B. $-CH_3$

C. –C≡C–

D. –Cl

O–H is polar with a lone pair on O (Chapter 3).

Q3 Recap

Above its pI, an amino acid carries a net:

**A. negative charge**

B. positive charge

C. zero charge

D. it depends on temperature

Chapter 7.

Q4 CH8-LO1

In chromatography, the phase that moves is called the:

**A. mobile phase**

B. stationary phase

C. solid phase

D. fixed phase

It carries the sample.

Q5 CH8-LO1

A compound that sticks strongly to the stationary phase travels faster.

True / False → **False**

It is held back, so it travels slower.

Q6 CH8-LO2

Silica gel is:

**A. polar**

B. nonpolar

C. a gas

D. an ion

Its surface is covered in –OH groups.

Q7 CH8-LO3

TLC stands for:

**A. thin layer chromatography**

B. total liquid chromatography

C. thermal layer chemistry

D. test for liquid content

A thin layer of adsorbent on a plate.

Q8 CH8-LO4

In gas chromatography, the sample must be:

**A. able to become a vapour without breaking down**

B. a solid salt

C. a protein

D. coloured

It is carried by a gas.

Q9 CH8-LO5

Which question does a pharmacy quality-control lab NOT need chromatography for?

**A. the price of the tablet**

B. the identity of the drug

C. the amount of drug

D. the impurities present

The other three are standard tests.

Q10 CH8-LO1

Have you seen ink or food colouring spread out on wet paper or tissue? Describe what happened and suggest why the colours separated.

The water climbed up the paper and carried the dyes with it. Each dye is attracted to the paper and to the water by different amounts, so some dyes moved further than others and the colours spread into separate bands. That is paper chromatography.

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